

LAB 2: Mission Briefing

In this lab, you will practice using **Python statements and expressions** to create a small interactive adventure program.

Learning Outcomes

Upon completion of this assignment, you will know how to use:

- Variables
- Assignment statements
- Arithmetic expressions
- String expressions
- Boolean expressions
- Input and output
- Type conversion
- Comments
- Operators

Scenario

You have been recruited for a **secret mission**.

Your program will ask the user several questions and use their answers to calculate information about their mission.

At the end, your program should display a **Mission Briefing** containing the results.

The program does **not** need if statements, loops, functions, or other control structures yet. The goal is to practice **statements and expressions only**.

Part 0 — REPO

In your local course repository, create a new folder called **lab02**.

Part 1 — Mission Information

Create a Python program named:

```
mission.py
```

Start the program with a comment containing your name, course, and assignment.

Your program should ask the user for:

- Their agent name
- Their age
- The number of years they have been training
- Their favorite color
- The number of gadgets they are carrying
- The number of minutes they have to complete the mission

Example:

```
Enter your agent name: Morgan
Enter your age: 24
How many years have you been training? 5
Enter your favorite color: blue
How many gadgets are you carrying? 4
How many minutes do you have? 37
```

Part 2 — Calculate Mission Statistics

Create variables and expressions that calculate the following.

1. Training Percentage

Calculate the percentage of the agent's life they have spent training.

For example:

```
5 years / 24 years × 1005 years / 24 years × 100
```

Store the result in:

```
training_percentagettraining_percentage
```

2. Gadget Density

Calculate the number of gadgets per year of training.

Store the result in:

```
gadget_densitygadget_density
```

3. Mission Seconds

Convert the number of mission minutes into seconds.

Store the result in:

```
mission_secondsmission_seconds
```

4. Mission Countdown

Assume the agent needs **7 minutes** to reach the mission location.

Calculate how many minutes remain for the actual mission.

Store the result in:

```
mission_time_remainingmission_time_remaining
```

Part 3 — Create a Mission Code

Create a variable called:

```
mission_codemission_code
```

The mission code should be created by combining:

- The agent's name
- Their favorite color
- Their age

For example:

```
Agent: Morgan  
Color: blue  
Age: 24Agent: Morgan  
Color: blue  
Age: 24
```

could produce:

```
MORGAN-BLUE-24MORGAN-BLUE-24
```

Hint: You can use string methods such as `.upper()`.

Part 4 — Boolean Expressions

Create Boolean expressions that answer these questions:

```
is_adult
has_many_gadgets
has_training_experienceis_adult
has_many_gadgets
has_training_experience
```

Use the following rules:

is_adult is True if the agent is at least 18.

has_many_gadgets is True if the agent has at least 5 gadgets.

has_training_experience is True if the agent has trained for more than 0 years.

For example:

```
is_adult = age >= 18is_adult = age >= 18
```

Part 5 — Mission Briefing

Use `print()` statements to produce a final mission briefing.

Your output should contain **all** of the following:

```
=====
SECRET MISSION BRIEFING
=====
```

```
Agent: Morgan
Mission Code: MORGAN-BLUE-24
```

```
Age: 24
Training: 5 years
Training Percentage: 20.83%
```

```
Gadgets: 4
Gadget Density: 0.80 per training year
```

```
Mission Time: 37 minutes
Mission Time Remaining: 30 minutes
Mission Time in Seconds: 2220
```

```
Adult Agent: True
Many Gadgets: False
Training Experience: True
```

```
=====
GOOD LUCK, AGENT!
=====
```

Your exact numbers will depend on the values entered by the user.

DELIVERABLE

Submit to Blackboard a link to the **mwcc-cs109/lab02** folder on GitHub. The folder should contain the **mission.py** file.